

# Python Programming Guide

## Introduction

Python is a high-level, interpreted, and object-oriented programming language created by Guido van Rossum in 1991. It is widely used in web development, artificial intelligence, machine learning, data science, automation, backend development, and cybersecurity.

Python is known for its simple syntax, readability, and powerful standard library.

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## Features of Python

- Easy to learn
  - Simple syntax
  - Open source
  - Cross-platform
  - Object-Oriented
  - Large community support
  - Huge collection of libraries
  - Automatic memory management
- 

## Variables

Variables are used to store data.

Example:

```
name = "Monika"
```

```
age = 21
```

```
cgpa = 8.9
```

```
print(name)
```

```
print(age)
```

```
print(cgpa)
```

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## **Data Types**

Python supports several built-in data types.

### **Integer**

```
age = 21
```

### **Float**

```
price = 99.99
```

### **String**

```
name = "Python"
```

### **Boolean**

```
is_student = True
```

### **List**

```
languages = ["Python", "Java", "C++"]
```

### **Tuple**

```
coordinates = (10,20)
```

### **Dictionary**

```
student = {  
    "name": "Monika",  
    "age": 21  
}
```

### **Set**

```
numbers = {1,2,3,4}
```

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## **Operators**

Python supports

- Arithmetic Operators
- Comparison Operators
- Assignment Operators
- Logical Operators

- Identity Operators
- Membership Operators

Example

```
a = 10
```

```
b = 5
```

```
print(a+b)
```

```
print(a-b)
```

```
print(a*b)
```

```
print(a/b)
```

```
print(a>b)
```

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## **Conditional Statements**

Example

```
marks = 85
```

```
if marks >= 90:
```

```
    print("Grade A")
```

```
elif marks >= 75:
```

```
    print("Grade B")
```

```
else:
```

```
    print("Grade C")
```

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## **Loops**

### **For Loop**

```
for i in range(5):
```

```
    print(i)
```

### **While Loop**

```
count = 1
```

```
while count <= 5:
```

```
    print(count)
```

```
    count += 1
```

---

## **Functions**

Functions help organize reusable code.

```
def add(a,b):
```

```
    return a+b
```

```
result = add(10,20)
```

```
print(result)
```

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## **Object-Oriented Programming**

Python supports OOP.

### **Class**

```
class Student:
```

```
    def __init__(self,name):
```

```
        self.name = name
```

```
    def display(self):
```

```
        print(self.name)
```

### **Object**

```
student = Student("Monika")
```

```
student.display()
```

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## Inheritance

```
class Animal:
```

```
    def sound(self):
```

```
        print("Animal Sound")
```

```
class Dog(Animal):
```

```
    def bark(self):
```

```
        print("Bark")
```

```
dog = Dog()
```

```
dog.sound()
```

```
dog.bark()
```

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## Exception Handling

```
try:
```

```
    number = 10/0
```

```
except ZeroDivisionError:
```

```
    print("Cannot divide by zero")
```

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## File Handling

Writing to a file

```
with open("data.txt","w") as file:
```

```
    file.write("Hello Python")
```

Reading a file

with open("data.txt","r") as file:

```
    print(file.read())
```

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## **Python Libraries**

Popular Python libraries include:

- NumPy
  - Pandas
  - Matplotlib
  - Scikit-learn
  - TensorFlow
  - PyTorch
  - Django
  - FastAPI
  - Flask
  - Requests
  - BeautifulSoup
  - LangChain
  - ChromaDB
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## **Applications of Python**

Python is widely used in:

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Data Science
- Automation
- Web Development
- Cybersecurity

- REST API Development
  - Cloud Computing
  - Game Development
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### **Advantages of Python**

- Beginner Friendly
  - Easy Syntax
  - Large Community
  - Huge Library Support
  - Platform Independent
  - Rapid Development
  - Excellent for AI and Backend Development
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### **Python Interview Questions**

#### **What is Python?**

Python is a high-level, interpreted programming language used for web development, AI, automation, and data science.

#### **What are Python's main features?**

- Simple syntax
- Open source
- Object-oriented
- Cross-platform
- Dynamic typing

#### **What is a list?**

A list is an ordered and mutable collection.

Example:

```
numbers = [1,2,3]
```

#### **What is a dictionary?**

A dictionary stores data as key-value pairs.

Example:

```
student = {  
    "name": "Monika",  
    "age": 21  
}
```

### **What is a function?**

A function is a reusable block of code.

Example:

```
def greet():  
    print("Hello")
```

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### **Summary**

Python is one of the most popular programming languages due to its simplicity, versatility, and extensive ecosystem. It is used in artificial intelligence, web development, automation, machine learning, cloud computing, and scientific computing. Python's readability and large community make it an excellent language for beginners and professionals alike.

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